MATH102BKK · HARBOUR.SPACE BANGKOK

Homework 4 — Function families

Session 4 · Polynomials, exponentials, logarithms, and trigonometric models

Set
Thu 10 Sep 2026
Due
Fri 11 Sep 2026, start of class

HOW TO ANSWER Every answer carries one sentence of justification. A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

BEFORE YOU START Conditions are part of an answer. Before using a logarithm, state which inputs are allowed. When you compare growth, say *for sufficiently large* n; one numerical example does not prove a growth claim.

Part A · Mechanics

1 POLYNOMIAL SHAPE

Let $p(x) = -3x^4 + 2x - 7$.

a. State the degree and leading coefficient of p.

b. Describe what the graph does at its left and right ends.

Give one sentence explaining why the lower-power terms do not change your answer to part (b).

SOLUTION The degree is 4 and the leading coefficient is -3. The degree is even and the leading coefficient is negative, so the graph goes down at both ends: $p(x) \to -\infty$ as $x \to -\infty$ and as $x \to \infty$. When $|x|$ is large, the leading term $-3x^4$ is much larger in size than $2x - 7$, so it determines the end behaviour.

2 QUADRATIC TRANSFORMATION

Start with $f(x) = x^2$. The function

$$g(x) = -(x-2)^2 + 4$$

is made from f.

a. Describe the shift and reflection.

b. Give the vertex and find the real roots of g.

SOLUTION Move the graph of $y = x^2$ right by 2, reflect it in the x-axis, then move it up by 4. Its vertex is $(2, 4)$. For roots, $-(x-2)^2 + 4 = 0 \iff (x-2)^2 = 4 \iff x - 2 = \pm 2$, so $x = 0$ or $x = 4$.

3 EXPONENTIAL EQUATION

Solve $5^{2x-1} = 125$. Rewrite 125 with base 5, then state why you may set the exponents equal.

> **SOLUTION** Since $125 = 5^3$, we have $5^{2x-1} = 5^3$. The base 5 is positive and is not 1, so the exponential function 5^t is one-to-one and $2x - 1 = 3$. Hence $x = 2$.

4 CHANGE OF BASE

Use the change-of-base formula to calculate $\log_4 32$. Show the formula you use and give the exact answer.

> **SOLUTION**
>
> $$\log_4 32 = \frac{\ln 32}{\ln 4} = \frac{\ln(2^5)}{\ln(2^2)} = \frac{5 \ln 2}{2 \ln 2} = \frac{5}{2}.$$
>
> The formula is valid because the input 32 is positive and the bases 4 and e are positive and not equal to 1.

Part B · Reasoning and modelling

5 LOGARITHMIC EQUATION

Solve over the real numbers:

$$\log_2(x-1) + \log_2(x+1) = 3.$$

State the domain before combining the logarithms. Check every possible answer in the original equation.

> **SOLUTION** Both logarithms require positive inputs: $x - 1 > 0$ and $x + 1 > 0$. Therefore $x > 1$. On this domain, the product law gives
>
> $$\log_2\big((x-1)(x+1)\big) = 3 \iff x^2 - 1 = 8 \iff x^2 = 9.$$
>
> The algebra gives $x = 3$ or $x = -3$, but the domain keeps only $x = 3$. Check: $\log_2(2) + \log_2(4) = 1 + 2 = 3$.

6 GROWTH COMPARISON

Put the following functions in order from slowest growth to fastest growth as the positive integer n becomes very large:

$$\log_2 n, \qquad n, \qquad n \log_2 n, \qquad n^2, \qquad 2^n.$$

In two or three sentences, explain why checking one value such as $n = 3$ does not prove the full ranking.

SOLUTION The eventual order is

$$\log_2 n < n < n \log_2 n < n^2 < 2^n.$$

This is an eventual statement: it becomes true once n is large enough, not necessarily at every positive integer. For example, at $n = 3$, $2^n = 8 < 9 = n^2$, but for sufficiently large n the exponential 2^n is larger than n^2.

7 PERIODIC MODEL

Let t be the number of hours after midnight. A simple model for temperature is

$$T(t) = 3 \sin\left(\frac{\pi}{6}(t-1)\right) + 20 \qquad (0 \le t \le 12).$$

a. State the amplitude, midline, period, and range.

b. At what time does the model first reach a maximum? At what time does it reach a minimum?

c. Explain why $(t-1)$ moves the graph right by 1 hour, rather than left by 1 hour.

SOLUTION The amplitude is 3, the midline is $y = 20$, the period is $\dfrac{2\pi}{\pi/6} = 12$ hours, and the range is $[17, 23]$. A maximum occurs when the sine input is $\pi/2$, so $\frac{\pi}{6}(t-1) = \frac{\pi}{2}$ gives $t = 4$. A minimum occurs when the input is $3\pi/2$, giving $t = 10$. In $f(t-1)$, the new input must be $t = 1$ to give the old value $f(0)$, so the graph moves right by 1.

Part C · Communicate precisely

8 FIND THE ERROR

A student writes: "$\log_2(u+v) = \log_2 u + \log_2 v$ because logarithms turn an operation into addition."

a. Identify the false statement.

b. Give a counterexample with positive u and v.

c. Write the correct logarithm law for a product, including its condition.

> **SOLUTION** The false statement is $\log_2(u+v) = \log_2 u + \log_2 v$. Take $u = v = 1$: $\log_2(1+1) = \log_2 2 = 1$, while $\log_2 1 + \log_2 1 = 0 + 0 = 0$. The correct product law is $\log_2(uv) = \log_2 u + \log_2 v$ for $u > 0$ and $v > 0$.

9 MAKE A CLAIM PRECISE

The statement "An exponential function is always bigger than a polynomial" is false or unclear as written. Rewrite it as a true statement about c^n and n^k, including all important conditions. Then give one numerical example showing why the words "for sufficiently large n " are necessary.

> **SOLUTION** One correct version is: "For every fixed $c > 1$ and $k > 0$, there is a positive integer N such that $c^n > n^k$ for every integer $n \geq N$." The word "fixed" matters: the bases and powers are not changing with n. For $c = 2$, $k = 2$, and $n = 3$, we have $2^3 = 8 < 9 = 3^2$, so the inequality is not true for every positive n.